

Chemistry, 2016 EC, entrance pilot test, booklet 018

Grade 12 · Chemistry · 2016 E.C.

Time allowed: 120 minutes

Questions: 67

Total marks: 67

Name _____ Grade _____ ID No. _____ Date _____

1. Among the following which one of the metal thin foil bombardments is CORRECT for the discovery neutrons? (1)
 - A. Gold with α -particle of radioactive substance.
 - B. Gold with β -particle of radioactive substance.
 - C. Beryllium with α -particle of radioactive substance.
 - D. Beryllium and gold with α -particle of radioactive substance.
2. The atomic mass of element ${}_1^1X$ and ${}_1^2Y$ are 1.6078 amu and 2.0141 amu, respectively. The relationship between element x and y is known as _____ (1)
 - A. isotopic
 - B. monomeric
 - C. isobaric
 - D. isotonic
3. The arrangement of electrons in the shells of an atom is known as _____ (1)
 - A. Electronegativity
 - B. Electro affinity
 - C. Electro positivity
 - D. Electronic configuration
4. What is the electronic configuration of sodium, Na? (Atomic number, Z = 11) (1)
 - A. $1s^2 2s^2 2p^5 3s^2$
 - B. $1s^2 2s^2 2p^6 3s^1$
 - C. $1s^2 2s^2 3p^6 3s^2$
 - D. $1s^2 2s^1 2p^6 3s^2$
5. Which periods in the periodic table are known as long periods? (1)
 - A. 1, 2 and 3
 - B. 1, 2, 3 and 4
 - C. 4, 5 and 6
 - D. 4, 5, 6 and 7
6. Which of the following is the CORRECT period and group number of an element with atomic number ${}_{20}^{40}Y$? (1)
 - A. Period 3 and group IVA
 - B. Period 3 and group IA
 - C. Period 4 and group VIIIA
 - D. Period 4 and group IIA
7. What is the force that holds two atoms or ions together in a compound? (1)
 - A. Chemical bond
 - B. Chemical equilibrium
 - C. Chemical reaction
 - D. Chemical kinetics
8. Which statement CORRECTLY explains the formation of an ionic bonding? (1)
 - A. Transfer of electrons from non-metal to metal.
 - B. Gaining of electrons between metals.
 - C. Transfer of electrons from metal to non-metal.
 - D. Gaining of electrons between non-metallic elements.
9. A bond between two atoms formed by sharing of one or more pairs of electrons is known as _____ (1)
 - A. ionic bond
 - B. hydrogen bond
 - C. covalent bond
 - D. metallic bond

-
10. Which of the following statements describes the properties of covalent compounds? (1)
- A. They conduct electric current in solution.
 - B. They have high melting and boiling points.
 - C. They are generally solid at ordinary temperature.
 - D. They are insoluble in polar solvents such as water.
-
11. Which of the following molecules has a significant dispersion force? (1)
- A. Ar
 - B. Xe
 - C. Kr
 - D. He
-
12. In the general convention of writing a chemical equation, reactant and product can be separated by? (1)
- A. point
 - B. dot
 - C. arrow
 - D. circle
-
13. How many grams of aluminum sulphate are formed when 27g of aluminum reacts with sulphuric acid? (At.wt. Al = 27 g, S = 32 g, H = 1 g and O = 16 g) (1)
- $\text{Al(s)} + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow \text{Al}_2(\text{SO}_4)_3 + \text{H}_2$
- A. 27 g
 - B. 54 g
 - C. 171 g
 - D. 342 g
-
14. Consider the following reaction: $2\text{KClO}_3 \rightarrow 2\text{KCl} + 3\text{O}_2$ (1)
- When 122.5 g of KClO_3 decomposes, 8.5 g of O_2 is formed. What is the percentage yield? (At.wt. K = 39 g, Cl = 35.5 g and O = 16 g)
- A. 1.77%
 - B. 7.77%
 - C. 17.7%
 - D. 77.7%
-
15. Which of the following statements CORRECTLY define oxidation state? It is the number of electrons that an atom (1)
- A. completely loses when it is combined with other atoms.
 - B. completely accepts when it is combined with other atoms.
 - C. equally shares electrons when it is combined with other atoms.
 - D. appears to have gained or lost when it is combined with other atoms.
-
16. Consider the following reaction: (1)
- $\text{Fe}_2\text{O}_3 + \text{CO} \rightarrow \text{Fe} + \text{CO}_2$
- Which substance is reduced and oxidized respectively?
- A. Iron and carbon
 - B. Carbon monoxide and iron
 - C. Carbon and iron
 - D. Iron and carbon monoxide
-
17. What is the term that describes a dynamic process in which the concentrations of reactants and products remain constant with time? (1)
- A. Chemical property
 - B. Chemical equilibrium
 - C. Chemical reaction
 - D. Chemical equation
-
18. What is the main reason for the gas and liquid NOT to have definite shape? They have: (1)
- A. higher intermolecular force
 - B. higher density and mass
 - C. lower intermolecular force
 - D. stronger force of attraction

19. What is the term that describes the phase change from solid to a gas state? (1)
- A. Deposition
B. Condensation
C. Fusion
D. Sublimation
20. What is the term used for naming groups attached to the compounds and are responsible for the differences in chemical properties of the compounds? (1)
- A. Functional groups
B. Organic groups
C. Inorganic groups
D. Alkane groups
21. What is the IUPAC name of the following organic compound? (1)
- [Figure: A chemical structural formula showing a horizontal six-carbon chain: $H_3C-C(CH_3)_2-CH(CH_3)-CH(CH_3)-CH_2-CH_3$. Specifically, the leftmost carbon is H_3C connected to a C atom with a CH_3 branch above and a CH_3 branch below; this C is connected to the next C which has a CH_3 branch above and an H below; this is connected to the next C with an H above and a CH_3 branch below; this is connected to the next C with an H above and an H below; which is finally bonded to $-CH_3$ on the far right.]*
- A. 3, 4, 5, 5-tetramethylhexane
B. 2, 2, 3, 4-tetramethylhexane
C. 2-ethyl-3, 4, 4-trimethylpentane
D. 4-ethyl-2, 2, 3-trimethylpentane
22. Which one of the following is CORRECT about the cathode in electrolysis? (1)
- A. Oxidation takes place at the cathode
B. Reduction takes place at the Cathode
C. lose of electrons occurs at the cathode
D. gain of electrons occurs at the anode
23. Which of the following natural resources is renewable? (1)
- A. petroleum
B. coal
C. soil
D. minerals
24. Among the following statements, which one CORRECTLY describes electrochemistry? It deals with the relationship between: (1)
- A. electrolytic cell and voltaic cell.
B. electrical energy and chemical energy.
C. electrical energy and mechanical energy.
D. biofuel conversion and electrical energy.
25. Consider the following Cell: (1)
- What is the direction of electron flow and the electrode where reduction takes place?
- [Figure: A diagram of an electrochemical cell consisting of two beakers. The left beaker contains a solution labeled "ZnSO₄ solution" and an electrode labeled "Zn electrode". The right beaker contains a solution labeled "CuSO₄ solution" and an electrode labeled "Cu electrode". A "salt bridge" connects the solutions in both beakers. An external wire connects the two electrodes through a "voltmeter".]*
- A. Zn to Cu, the Zn electrode
B. Cu to Zn, the Zn electrode
C. Cu to Zn, the Cu electrode
D. Zn to Cu, the Cu electrode
26. A diagnostic laboratory reported $6.51 \times 10^3 \mu\text{g}$ blood sample analysis of chromium. What is the value of the report in gram? (1)
- A. 0.00651 g
B. 0.0651 g
C. 0.651 g
D. 651 g

27. How many decimal places are there in 0.00012? (1)

- A. 2
B. 4
C. 5
D. 6

28. How many significant figures are there in 0.01698g? (1)

- A. 4
B. 5
C. 6
D. 7

29. What is the term used to express the spontaneous emission of radiation from the unstable nuclei of certain isotopes? (1)

- A.** Radioactivity **B.** Radio-isotope
C. Radioactive decay **D.** Radiocarbon dating

30. Isotopes $^{63}_{29}\text{Cu}$ (62.9298 amu) accounts for 69.09% and isotopes $^{65}_{29}\text{Cu}$ (64.9278 amu) accounts for 30.91%. What is the average atomic mass of copper? (1)

- A. 61.9289 amu
B. 63.5409 amu
C. 63.9298 amu
D. 64.5409 amu

31. The threshold frequency of a metal surface is in the violet light region. What types of radiation will cause photoelectrons to be emitted from the metal surface? (1)

- A. IR radiation
B. Red light
C. UV radiation
D. Radio wave

32. Which quantum number describes the shape of an orbital? (1)

- A. The magnetic number B. The principal number
C. The spin number D. The azimuthal number

33. What is the shape of s orbital? (1)

- A. spherical
B. dumb bell
C. complex
D. distorted dumb bell

34. For the polyatomic ion, CO_3^{2-} , which one of the following CORRECTLY represents the resonance structure(s)? (1)

[Figure: Four sets of Lewis resonance structures for the carbonate ion, CO_3^{2-} , enclosed in square brackets with a 2^- charge: - A shows two structures connected by a double-headed resonance arrow: the first has a double bond to the top oxygen and single bonds to the two lower oxygens; the second has a double bond to the bottom-left oxygen and single bonds to the top and bottom-right oxygens. - B shows two structures connected by a double-headed resonance arrow: both have a double bond to the top oxygen and single bonds to the two lower oxygens. - C shows three structures connected by double-headed resonance arrows: the first has a double bond to the top oxygen (with two lone pairs) and single bonds to the two lower oxygens (each with three lone pairs); the second has a double bond to the bottom-left oxygen and single bonds to the top and bottom-right oxygens; the third has a double bond to the bottom-right oxygen and single bonds to the top and bottom-left oxygens. - D shows three structures connected by double-headed resonance arrows, each having only single C-O bonds to all three oxygen atoms.]

- A. Two resonance structures showing a C=O double bond alternating between the top oxygen and the bottom-left oxygen
- B. Two resonance structures showing a C=O double bond to the top oxygen in both structures
- C. Three resonance structures showing a C=O double bond alternating among the top oxygen, the bottom-left oxygen, and the bottom-right oxygen
- D. Three structures showing only single bonds between the central carbon and each of the three oxygen atoms

35. Which of the following molecule has a polar bond and is a non-polar molecule? (1)
- A. HCl
B. CHCl₃
C. CO₂
D. O₂
36. What intermolecular force occurs in compounds that contain H-F, H-O and H-N bonds? (1)
- A. London force
B. Dipole-dipole
C. Induced dipole
D. Hydrogen bond
37. Which one of the following statements CORRECTLY describes hybridization? (1)
- A. It is the mixing of atomic orbitals to produce a new set of orbitals.
B. Sideway overlapping of atomic orbitals produces sigma bonding.
C. Combination of one s and one p orbitals produces a sp² hybrid orbital.
D. The new hybrid orbitals is higher than the total number of atomic orbitals combined.
38. Which of the following statement is CORRECT about the molecular orbital theory? (1)
- A. When two pure p-orbitals overlap end-to-end, they form two pi-molecular orbitals.
B. Bonding molecular orbitals have a region of high electron density between the nuclei.
C. Antibonding molecular orbitals have a region of high electron density between the nuclei.
D. The number of molecular orbitals formed is higher than the number of atomic orbitals combined.
39. Consider the following model of sodium chloride: (1)
- What is the expected crystalline structure of the sodium chloride crystal?
[Figure: A 3D ball-and-stick model of a sodium chloride crystal lattice. A legend to the lower left indicates that a larger shaded circle represents Cl⁻ and a smaller shaded circle represents Na⁺. The lattice consists of alternating Na⁺ and Cl⁻ ions forming a cubic lattice. A bracket below one repeat unit at the base is labeled "Unit cell".]
- A. Molecular crystal
B. Covalent network crystal
C. Metallic crystal
D. Ionic crystal
40. At what condition do the forward and reverse reactions are taking place at the same rate? It is in state of _____. (1)
- A. dynamic equilibrium
B. equilibrium constant
C. phase equilibrium
D. chemical kinetics
41. The ratio of the concentration of products to the concentration of reactants raised to the power of their coefficient is known as _____. (1)
- A. reaction rate
B. reaction quotient
C. reaction intermediate
D. reaction mechanism
42. Consider the following reaction: $2\text{CO (g)} + \text{O}_2 \text{ (g)} \rightleftharpoons 2\text{CO}_2 \text{ (g)}$ (1)
- Has an equilibrium constant $K_c = 4.3 \times 10^6$, [CO] = 0.10M, [CO₂] = 10M and [O₂] = 0.1M
To which direction will the reaction proceed to attain equilibrium?
- A. Backward
B. to the left
C. Forward
D. to reverse
43. Consider the following reaction: $\text{N}_{2(g)} + 3\text{H}_{2(g)} \rightleftharpoons 2\text{NH}_{3(g)}$ $\Delta H = -92 \text{ kJ/mol}$ (1)
- What happens when the temperature of the system is increased?
- A. More NH₃ is formed
B. Less N₂ is formed
C. Equilibrium shifts to the left
D. Equilibrium shifts to the right

44. Which of the following is a chemical property of esters? (1)

- | | |
|---------------|---------------|
| A. Hydrolysis | B. Solubility |
| C. Volatility | D. Odour |

45. What is the name of the compounds formed when carboxylic acids react with alcohol? (1)

- | | |
|-----------|------------|
| A. Ethers | B. Ketones |
| C. Esters | D. Amides |

46. Which one of the following structures represents a triglyceride? (1)

[Figure: Four chemical structures are shown as choices:

A shows a glycerol backbone: H_2C-OH bonded vertically to $HC-OH$, which is bonded vertically to H_2C-OH .

B shows a three-carbon chain where each carbon is bonded to an acyl/ester group: $H_2C-COOR_1$ bonded vertically to $HC-COOR_2$, which is bonded vertically to $H_2C-COOR_3$.

C shows a three-carbon chain where each carbon is bonded to an ether group: H_2C-OR_1 bonded vertically to $HC-OR_2$, which is bonded vertically to H_2C-OR_3 .

D shows a two-carbon chain: H_2C-OR_1 bonded vertically to H_2C-OR_2 .]

- | | |
|--|--|
| A. H_2C-OH

$HC-OH$

H_2C-OH | B. $H_2C-COOR_1$

$HC-COOR_2$

$H_2C-COOR_3$ |
| C. H_2C-OR_1

$HC-OR_2$

H_2C-OR_3 | D. H_2C-OR_1

H_2C-OR_2 |

47. Which of the following reaction represents the ionization of acid according to Arrhenius definition? (1)

- A. $NaCl(s) + H_2O(l) \rightleftharpoons Na^+(aq) + Cl^-(aq)$
B. $NH_3(aq) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$
C. $OCl^-(aq) + H_2O(l) \rightleftharpoons HOCl(aq) + OH^-(aq)$
D. $HClO_2(aq) + H_2O(l) \rightleftharpoons ClO_2^-(aq) + H_3O^+(aq)$

48. How is acid named according to the Lewis acid-base concept? (1)

- | | |
|----------------------|------------------------------|
| A. a proton donor | B. an electron pair donor |
| C. a proton acceptor | D. an electron pair acceptor |

49. The ion product constant for water can be written as $K_w = [H_3O^+][OH^-]$. What happens to the solution when $[H_3O^+] > [OH^-]$? (1)

- | | |
|---------------------------------|----------------------------------|
| A. The solution becomes neutral | B. The solution becomes basic |
| C. The solution becomes acidic | D. The pH is higher than the pOH |

50. Which of the following given pairs represents an example of buffer system? (1)

- | | |
|---|---|
| A. $NaCl/H_2O$ and $H_2O/NaOH$ | B. $LiOH/HNO_3$ and $NaOH/NH_4Cl$ |
| C. NH_4OH/NH_4Cl and CH_3COOH/CH_3COONa | D. NH_4OH/CH_3COOH and NH_4Cl/CH_3COONa |

51. Which one of the following is TRUE about oxidation? Oxidation is: (1)

- | | |
|------------------------|---------------------------------|
| A. loss of electrons | B. gain of electrons |
| C. sharing of electron | D. decrease in oxidation number |

52. Which ion is preferentially discharged at the anode during the electrolysis of aqueous solution of dilute H_2SO_4 ? (1)

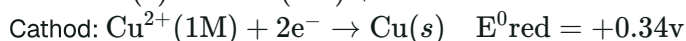
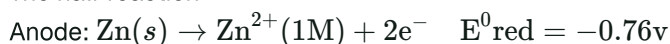
- A. SO_4^{2-} B. OH^-
C. H^+ D. Na^+

53. Which one of the following statements describes Faraday's second law of electrolysis? (1)

- A. The masses of different substance liberated by the same amount of electricity are proportional to their equivalent masses
- B. The rate of reaction is directly proportional to the product of the concentration raised to the power of their respective coefficients
- C. The mass of a substance deposited or dissolved at electrodes is proportional to the quantity of electricity passed through the electrolyte
- D. The electromotive force of attraction between atoms is proportional to the product of the two charges and inversely proportional to the square of their radius.

54. Consider the following Galvanic cell: (1)

The half reaction



What is the cell potential of the voltaic cell at 25°C ?

[Figure: A diagram of a galvanic cell with two beakers connected by an inverted U-tube salt bridge labeled " $\text{KNO}_3(\text{aq})$ Salt bridge". The left beaker contains " $1.00 \text{ M Zn}(\text{NO}_3)_2(\text{aq})$ " solution and a strip labeled "Zn" and "Anode (-)". The right beaker contains " $1.00 \text{ M Cu}(\text{NO}_3)_2(\text{aq})$ " solution and a strip labeled "Cu" and "Cathode (+)". An external wire connects the Zn and Cu electrodes through a "Voltmeter", with arrows showing electron flow from the Zn electrode to the Cu electrode.]

- A. 0.34v
B. 0.76v
C. 1.10v
D. 1.42v

55. Among the following, which one is the negative effect of corrosion of metals? (1)

- A.** Strengthening of the metal
- B.** Increasing the conductivity of the metals
- C.** Giving lustrous or shining appearance to the metals
- D.** Damaging the metals by converting them to their oxides

56. Which of the following is a well-organized facility with high degree of automation and provide goods and services? (1)

- A.** Industry **B.** Polymerization
C. Calcination **D.** Densification

57. Consider the following schematic diagram: (1)

The diagram above displays the production of sulphuric acid in the contact processes, which step uses V_2O_5 catalyst with very high temperature?

[Figure: A schematic diagram showing the contact process for sulphuric acid production, separated horizontally into three stages: 'First Step', 'Second Step', and 'Third Step'. In the 'First Step', 'sulphur' from above and 'Dry Air' from below enter a 'Furnace' to produce ' SO_2 '. In the 'Second Step', the gas pipe labelled ' $\text{SO}_2 + \text{O}_2$ ' leads into a tall reaction vessel, which outputs ' SO_3 '. In the 'Third Step', 'Concentrated sulphuric Acid' is added to ' SO_3 ' in an absorber tower to produce ' $\text{H}_2\text{S}_2\text{O}_7$ (Oleum)'. Adding 'Water' to the oleum produces ' H_2SO_4 ', with an exhaust 'Pipe' leading to an emission stack.]

- A.** Third step
- B.** First step
- C.** Second step
- D.** First and second steps

-
- 58.** Which one of the following increases the production of nitric acid in the Ostwald's process? (1)
- | | |
|----------------------------|------------------------|
| A. Using V_2O_5 catalyst | B. Oxidation of NH_3 |
| C. Oxidation of NO_2 | D. Removal of H_2O |
-
- 59.** Which one of the following is CORRECT about the uses of animal hide? It is used for: (1)
- | | |
|------------------------------------|---|
| A. making soap and detergents | B. making garments and handbags |
| C. making kitchenware and bathroom | D. removing impurities from waste water |
-
- 60.** Among the following, which one is NOT a natural condensation polymer? (1)
- | | |
|--------------|--------------|
| A. Starch | B. Proteins |
| C. Cellulose | D. Polyester |
-
- 61.** Which one of the following materials is capable of conducting electricity without transfer of matter? (1)
- | | |
|-----------------|-------------|
| A. Electrolytic | B. Ionic |
| C. Metallic | D. Covalent |
-
- 62.** What is the point of conversion in a titration where an indicator changes its color? It is: (1)
- | | |
|---------------------|----------------------|
| A. equivalent point | B. endpoint |
| C. minimum point | D. equilibrium point |
-
- 63.** Which of the following statement is INCORRECT about polysaccharide? They are (1)
- | | |
|------------------------------|----------------------------------|
| A. made by addition reaction | B. called macromolecules |
| C. called biopolymers | D. made by condensation reaction |
-
- 64.** What is the technique used to prevent pollution caused by over use of fertilizers? (1)
- | |
|--|
| A. Excessive use of Inorganic fertilizer |
| B. Use of urea and phosphate fertilizers |
| C. Excessive use of non-biodegradable wastes |
| D. Use of natural and easily degradable wastes |
-
- 65.** Which one of the following gases has higher contribution to global warming? (1)
- | | |
|-------------------|--------------------------|
| A. Methane | B. Nitrous oxide |
| C. Carbon dioxide | D. Halogenated compounds |
-
- 66.** Among the following, which one is the cause of soil pollution? (1)
- | |
|---|
| A. Reduction of forest fires and smoking |
| B. Excessive use of synthetic fertilizers |
| C. Avoiding using of products with chemicals |
| D. Recycling industrial and agricultural wastes |

67. Which of the following represents the structure of disaccharide?

(1)

[Figure: Four chemical structures labeled A, B, C, and D showing cyclic representations of carbohydrates: - Option A shows two six-membered pyranose rings connected by an ether (glycosidic) linkage between the anomeric carbon of one ring and a carbon of the second ring, representing a disaccharide. - Option B shows a single pyranose ring with substituents $\text{HOH}_2\text{C}-$, $\text{HO}-$, and $-\text{OH}$, representing a monosaccharide. - Option C shows four pyranose rings linked linearly via glycosidic bonds with wavy bonds extending at both ends, representing a polysaccharide chain. - Option D shows four pyranose rings connected in a branched structure via glycosidic bonds.]

- A. [Structure showing two six-membered cyclic sugar units linked together by a glycosidic oxygen bond]
- B. [Structure showing a single six-membered cyclic sugar unit]
- C. [Structure showing a linear chain of four repeating six-membered cyclic sugar units with wavy bonds indicating a continuing polymer chain]
- D. [Structure showing a branched chain of four six-membered cyclic sugar units]

— END OF EXAMINATION —

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